

Data Analysis, Visualization, and Communication

Laboratórios de Sistemas e Serviços

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2026

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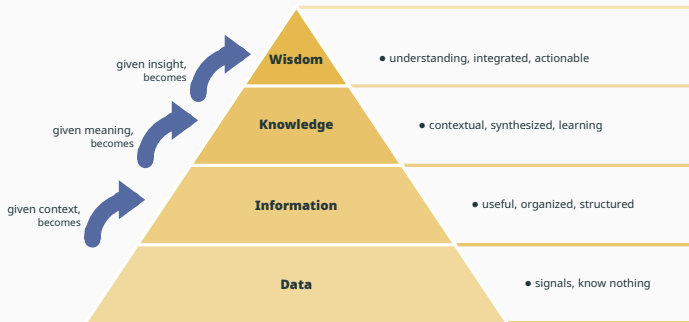
Summary

From Data to Knowledge

The journey from raw data to actionable knowledge.

- **Data:** Discrete, objective facts or observations.
- **Information:** Data that is processed to be useful; answers “Who, What, Where, When”.
- **Knowledge:** Application of data and information; answers “How”.
- **Wisdom:** Evaluated understanding; answers “Why”.

The DIKW Pyramid



- As we move up the pyramid, the **meaning** and **value** increase.
- As engineers, we build systems that automate this transformation.

The Data Science Pipeline I

A standard workflow for taking advantage of data.

1. **Collection:** Gathering data from sensors, APIs, or databases.
2. **Cleaning:** Handling missing values, noise, and outliers.
3. **Exploration (EDA):** Understanding the statistical properties of the data.
4. **Analysis:** Building models or performing complex queries.
5. **Visualization:** Communicating the results effectively.

The Data Science Pipeline II

- **Iteration:** The pipeline is not linear; you often go back to cleaning after exploration.
- **Automation:** Use scripts (Python) and pipelines to make this repeatable.
- **Quality:** “Garbage in, garbage out” – the cleaning phase is the most critical.

Data Loading & Manipulation

The DataFrame Concept I

A DataFrame is the central data structure in modern data science.

- **Conceptual Model:** An in-memory spreadsheet or SQL table.
- **Axis 0:** Rows (Samples/Observations).
- **Axis 1:** Columns (Variables/Features).
- **Index:** Unique labels for identifying rows.

The DataFrame Concept II

- **Homogeneity:** Each column usually contains a single data type (Int, Float, String).
- **Alignment:** Data is automatically aligned based on labels during operations.
- **Libraries:** Used by Pandas (Python), Polars (Python/Rust), and Spark (Big Data).

Tools: Pandas vs. Polars

Feature	Pandas	Polars
Backend	Python / C	Rust
Threading	Single-threaded	Multi-threaded
Evaluation	Eager	Lazy
Memory	High copies	Efficient (Arrow)

- Use **Pandas** for smaller datasets and compatibility.
- Use **Polars** for performance and massive datasets.

Data Cleaning

Missing Values (Imputation)

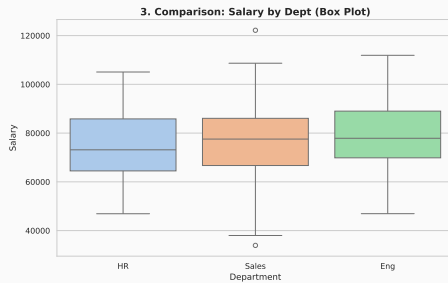
Real-world data is messy and often contains holes (NaN, Null).

- **Drop:** Remove rows with any missing values.
- **Fill (Constant):** Replace with zero or a default string.
- **Fill (Statistical):** Replace with Mean, Median, or Mode.
- **Mean:** Good for normal distributions.
- **Median:** Better when outliers are present.

Outlier Detection with IQR

Outliers are extreme values that can skew your analysis.

- **The Box Plot Method:** Uses quartiles to define “normal” range.
- **IQR:** $Q3 - Q1$.
- **Lower Bound:** $Q1 - 1.5 \times IQR$
- **Upper Bound:** $Q3 + 1.5 \times IQR$



Box Plot Outliers

Making sure all variables have a similar range.

- **Normalization:** Scale data to $[0, 1]$.

■

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

- **Standardization:** Center data around Mean=0 with Std Dev=1.

■

$$Z = \frac{X - \mu}{\sigma}$$

Exploratory Data Analysis (EDA)

Summarizing data with a few numbers.

- **Mean / Median / Mode:** Central tendency.
- **Variance / Std Dev:** Dispersion.
- **Quantiles:** 25%, 50%, 75% markers.

Correlation (Pearson)

Measuring the linear relationship between two variables.

- **Range:** $[-1, +1]$.
- **+1:** Perfect positive linear relationship.
- **-1:** Perfect negative linear relationship.
- **0:** No linear relationship.
- **Warning:** "Correlation does not imply causation."

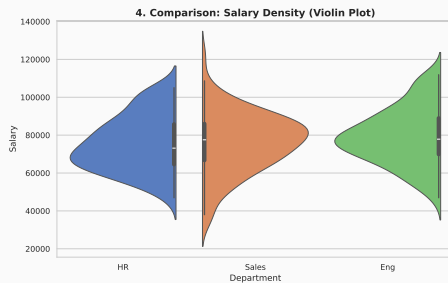
Data Visualization

Human brains are optimized for pattern recognition in images.

- **Matplotlib:** The low-level engine; infinite control.
- **Seaborn:** High-level statistical visualization; beautiful defaults.

Choosing the Right Plot

- **Distribution:** Histogram or KDE.
- **Comparison:** Bar Chart or Box Plot.
- **Relationship:** Scatter Plot or Line Plot.
- **Density:** Violin Plot (Box Plot + KDE).



Plot Comparison

Exporting: Raster vs. Vector

- **Raster (.PNG, .JPG):** Grid of pixels. Quality drops when zoomed. Good for web.
- **Vector (.PDF, .SVG):** Mathematical paths. Infinite scalability. Essential for **Academic Papers**.

Data Communication & Sharing

Documentation and APIs are how we share knowledge.

- **Markdown:** For documentation (READMEs).
- **LaTeX:** For formal academic reports.
- **Jupyter:** For interactive reports (Notebooks).

Sharing Data: Serialization for Exchange

- **Serialization:** Converting an object into a stream of bytes (JSON/Protobuf).
- **Schemas:** A contract (agreement) between producer and consumer.
- **Metadata:** Units, source, and timestamps (ISO 8601).

- **REST APIs:** The standard for web services (HTTP GET/POST + JSON).
- **MQTT:** Lightweight protocol for IoT (Publish/Subscribe).
- **WebSockets:** Real-time, full-duplex communication.

Summary

Summary

- **DIKW:** Data is the raw material; Knowledge is the goal.
- **Pipeline:** Cleaning and EDA are the most critical phases.
- **Tools:** Master Pandas/Polars for analysis and Seaborn for visualization.
- **Sharing:** Use standard formats, schemas, and APIs to communicate results.
- **Jupyter:** The laboratory for experimentation and storytelling.